

Energy consumption for business travel at KfW Group¹ in megawatt-hours

	2014	2015	2016	2017	2018
Car (petrol and natural gas)	200	263	659	416	428
Car (diesel)	6,725	7,391	5,071	5,953	5,506
Train	968	944	1,095	1,248	1,167
Flights (Germany)	5,027	4,409	4,839	4,626	4,785
Flights (Europe)	2,937	2,374	2,172	2,587	2,481
Flights (international)	25,850	19,809	19,558	22,235	23,799
Total	41,707	35,190	33,394	37,065	38,166
Per capita	6.6	5.5	5.0	5.0 ²	5.0

¹ Calculated according to figures from the Association for Environmental Management and Sustainability in Banks, Savings Banks and Insurance Companies (VfU); factors including upstream chain

² The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Measures for reducing energy consumption

In the context of a project for the energy-efficient restructuring of the central heating and cooling systems at the Frankfurt site, KfW completely restructured the system between the end of 2016 and spring 2019. This project was a response to the changing consumption structures within the building complex and to the challenges of the energy transition.

The measure was and is aimed at establishing a centralised, flexible and energy-efficient energy supply to the location that is sustainable in the long term. The solution uses large-scale heat pumps and combined heat and power systems to supply heating. Furthermore, waste heat from the data centre in the Westarkade building in Frankfurt will be used to heat office space in the future. The new heat pumps will raise the low temperature in the data centre to a generally usable level,

enabling it to supply a portion of the entire heating network. In respect of cooling, the company launched a highly efficient turbocore chiller in 2018. The combined heat and power system has been running on a trial basis since the end of the year under review. Altogether, these measures will enable carbon emissions to drop by up to 1,500 tonnes per year.

In 2017/2018, part of the existing uninterruptible power supply (UPS) system was exchanged for modern, smaller models in an additional project. Efficiency gains of 220 megawatt-hours of electricity per year are achieved as a result of the reduction in power loss.



In the future, the Frankfurt site will be able to save up to 1,500 tonnes of CO₂ each year

Emissions

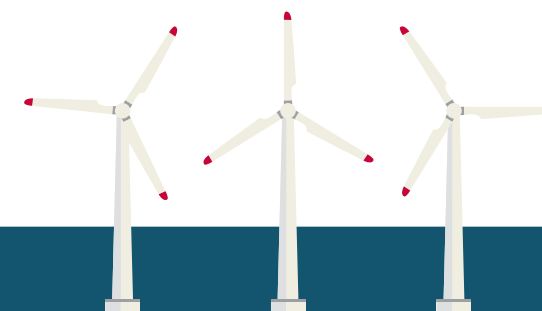
KfW Group's banking operations generate carbon emissions as a result of the consumption of energy and heat, the use of fuel during business trips, and due to events. The goal is to ensure climate-neutral banking operations wherever possible.

Climate-neutral banking operations

KfW Group's banking operations are largely climate-neutral in the areas of energy (buildings) and business trips. KfW offsets the emissions that remain despite the conversion to renewable energy sources and the use of more environmentally friendly transportation. Furthermore, it also assigns a CO₂ factor to the total green energy it consumes, as the use of green power still generates emissions. Electricity consumption at the external data centre is also taken into account in the compensation process. So far, not all avoidable emissions have been completely offset. KfW Group continually strives to record and offset as many emissions as possible.

To offset its carbon emissions, KfW Group purchases high-value Clean Development Mechanism (CDM) certificates on the market as well and discontinues them for good. In 2018, the majority of these certificates originated from a wind power project in India.

When evaluating its carbon equivalents (CO₂e), KfW uses international standards set by the Greenhouse Gas Protocol, and emissions are divided into Scopes 1 to 3. Scope 1 fell marginally by 0.8%. The rises under Scope 3 were caused by increased travel and events. The unusual 38.7% increase in Scope 2 was the result of increased use of heating oil and diesel-powered emergency generators due to building work. Nevertheless, the amount of CO₂e emitted per capita remained constant overall.



In 2018, CO₂ emissions were primarily offset by certificates for a wind power project in India

CO₂e emissions of KfW Group¹ in tonnes

	2014	2015	2016	2017	2018
Emissions from direct energy consumption (Scope 1) ²	4,636	5,095	5,260	5,616	5,571
Emissions from indirect energy consumption (Scope 2) ³	2,073	1,097	1,400	1,368	1,898
Total business travel (Scope 3) ⁴	9,446	7,394	7,411	8,208	8,665
Events (Scope 3) ⁵	102	102	74	87	101
Total	16,257	13,628	14,145	15,278	16,236
Per capita	2.6	2.2	2.1	2.1 ⁶	2.1 ⁶

¹ Unavoidable CO₂e emissions at the KfW Group have been offset since 2006.

² Natural gas, wood pellets and own fleet

³ Green electricity, district heating, oil and emergency power generators (diesel)

⁴ Only flights; short and long-distance rail travel is CO₂e-neutral per se.

⁵ Emissions for participants' travel to and from KfW buildings for events

⁶ The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Location- and market-based CO₂ emissions of KfW Group according to Scope 2 (2017) in tonnes

	Location-based, 2017	Market-based, 2017	Location-based, 2018	Market-based, 2018
Green electricity	5,869	4,636	5,819	4,596
Emergency power generators (diesel)	2,555	2,073	3,577	2,902
Heating oil	8,113	9,446	80	80
District heating	55	102	88	163
Total	16,592	16,257	9,564	7,732

According to the Greenhouse Gas (GHG) Protocol, we identified the Scope 2 emissions (from green electricity, district heating, oil and diesel emergency power generators) according to two different factors: location-based (average factor for Germany) and market-based (factor for the energy supplier) factors.

This data has been recorded since 2017.

Resource consumption

Paper is one of the essential materials of a financial institution. KfW Group continuously strives to reduce consumption. This is achieved, for example, by increasing employee awareness of the environmental impacts of paper consumption and

by switching from paper-based to digital processes in specific areas. Despite an increasing level of digitisation at KfW, the total amount of paper consumed increased by around 9% in the year under review. Per capita consumption rose by around 2 kilograms in total.

By far, the largest percentage of paper used at KfW Group – nearly 84% – is recycled paper with the “Blue Angel” environmental label. Publications intended for public use are printed exclusively on FSC® paper, which means that we do not use any ECF, TCF or fresh-fibre paper bleached with elemental chlorine. Furthermore, the group only works with climate-neutral printing companies – a further contribution towards climate neutrality at KfW.

84%

of KfW Group's office
paper bears the
“Blue Angel” label

KfW Group paper consumption in Germany in kilograms

	2014	2015	2016	2017	2018
Recycled paper (Blue Angel) ¹	308,385	251,150	225,863	196,084	213,000
Writing paper (recycled paper – Blue Angel) ¹	5,260	3,351	3,171	2,635	1,865
Virgin fibre paper for promotional materials (FSC mixed sources label)	72,931	153,512	73,291	32,528	39,704
Total	386,576	408,413	302,325	231,247	254,569
Per capita	61.5	64.3	45.5	31.2 ²	33.3
of which, recycled paper (in %)	81	62	76	86	84

¹ Refers to 80 g/m² A4 printer and copy paper.

² The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Disclosures	Page	Omission/comment
GRI 205: Anti-corruption 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 30-33	
GRI 205-1: Operations assessed for risks related to corruption	p. 32, 97	
GRI 205-2: Communication and training about anti-corruption policies and procedures	p. 31, 32	
GRI 205-3: Confirmed incidents of corruption and actions taken		In the year under review, nine cases of fraud and/or corruption were reported by KfW.
GRI 206: Anti-Competitive Behaviour 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)		KfW itself is a public promotional bank that promotes the economy, society and environment in Germany, Europe and the world. It operates in ways that are competitively neutral and therefore does not intervene in banking competition. The promotional mandate determined by the state is anchored in the KfW Law. KfW IPEX-Bank GmbH – a wholly owned subsidiary of KfW – is an independent financial institution operating on commercial principles in the market. DEG exclusively and directly pursues non-profit objectives.
GRI 206-1: Legal actions for anti-competitive behaviour, anti-trust, and monopoly practices		See Management Approach 206; no legal proceedings related to anti-competitive conduct have been filed against KfW IPEX-Bank.
GRI 300: Environment		
GRI 301: Materials 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 85, 92	
GRI 301-1: Materials used by weight or volume	p. 92	
GRI 301-2: Recycled input materials used	p. 92	
GRI 301-3: Reclaimed products and their packaging materials		Not relevant because KfW Group does not produce anything with such packaging material.
GRI 302: Energy 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 85, 86	
GRI 302-1: Energy consumption within the organisation	p. 87	
GRI 302-2: Energy consumption outside of the organisation	p. 88, 89	
GRI 302-3: Energy intensity	p. 87-89	
GRI 302-4: Reduction of energy consumption	p. 89	
GRI 302-5: Reductions in energy requirements of products and services		It is not possible to demonstrate the reduction in energy consumption per product or service in a bank.

Disclosures	Page	Omission/comment
GRI 303: Water 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 85, 93	
GRI 303-1: Water withdrawal by source	p. 93	
GRI 303-2: Water sources significantly affected by withdrawal of water		No groundwater is removed from protected areas.
GRI 303-3: Water recycled and reused	p. 93	KfW processes neither waste water nor sewage itself.
GRI 304: Biodiversity 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 49, 86, 105	
GRI 304-1: Operational sites owned, leased, managed in, or adjacent to, protected areas and areas of high biodiversity value outside protected areas		For a financial institution such as KfW Group, this indicator is irrelevant.
GRI 304-2: Significant impacts of activities, products, and services on biodiversity	p. 47, 86, 105	
GRI 304-3: Habitats protected or restored		For a financial institution such as KfW Group, this indicator is irrelevant.
GRI 304-4: IUCN Red List species and national conservation list species with habitats in areas affected by operations		For a financial institution such as KfW Group, this indicator is irrelevant.
GRI 305: Emissions 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 85, 90	
GRI 305-1 Direct (Scope 1) GHG emissions	p. 90, 91	
GRI 305-2: Energy indirect (Scope 2) GHG emissions	p. 90, 91	
GRI 305-3: Other indirect (Scope 3) GHG emissions	p. 90, 91	
GRI 305-4: GHG emissions intensity	p. 90, 91	
GRI 305-5: Reduction of GHG emissions	p. 90	
GRI 305-6: Emissions of ozone-depleting substances (ODS)		None.
GRI 305-7: Nitrogen oxides (NOx), sulfur oxides (SOx), and other significant air emissions		The combustion installations used by KfW are subject to annual statutory inspections. The measurement and recording of NOx and SOx readings is not required by law due to the building size and due to the fuels used (natural gas and wood pellets).

Disclosures	Page	Omission/comment
GRI 306: Effluents and Waste 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 85, 94, 95	
GRI 306-1: Water discharge by quality and destination	p. 94	
GRI 306-2: Waste by type and disposal method	p. 95, 96	
GRI 306-3: Significant spills		None.
GRI 306-4: Transport of hazardous waste		Due to the relatively small amount, KfW does not distinguish between hazardous waste in terms of the type of transportation and treatment.
GRI 306-5: Water bodies affected by water discharges and/or runoff		None.
GRI 307: Environmental Compliance 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 31	
GRI 307-1: Non-compliance with environmental laws and regulations		None.
GRI 308: Supplier Environmental Assessment 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 96, 97	
GRI 308-1: New suppliers that were screened using environmental criteria		Because the definition and, as a result, the group of suppliers which were surveyed on environmental criteria changed in 2019, a response is not required.
GRI 308-2: Negative environmental impacts in the supply chain and actions taken	p. 97	
GRI 400: Social		
GRI 401: Employment 2016		
GRI 103: Management Approach 2016 (including 103-1, 103-2, 103-3)	p. 58	
GRI 401-1: New employee hires and employee turnover	p. 58, 79, 80	Details about employee turnover and new employee hires according to region are not surveyed. The key figures are not relevant for management as the majority of our employees are in Germany. Turnover by age group is also not recorded at KfW Group.
GRI 401-2: Benefits provided to full-time employees that are not provided to temporary or part-time employees	p. 59-61	
GRI 401-3: Parental leave	p. 81	